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Uniform Sampling of Self-Orthogonal Codes via Orthogonal Group Actions

and codes with prescribed hull dimension — journal version, arXiv:2611.03578

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arXiv:2611.03578

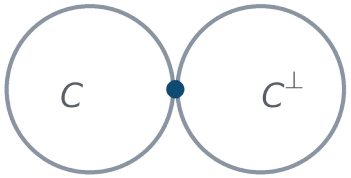
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The hull of a linear code

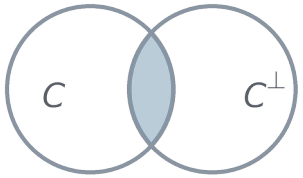
Hull(C) = C ∩ $C^{\perp}$ — three inner products, three hulls

C ≤ V an [n, k] code, $C^{\perp}$ its orthogonal space for some inner product (a non-degenerate reflexive form) B.

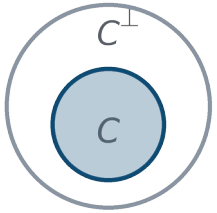
Hull(C) := C ∩ $C^{\perp}$, ℓ := dim Hull(C) ∈ {0, …, k}



ℓ = 0 · LCD
hull trivial



0 < ℓ < k
generic



ℓ = k · self-orthogonal
hull = C

A non-degenerate reflexive form is symmetric, alternating or Hermitian (up to a scalar) — so there are **exactly three hulls**:

Inner product	B(x, y)	Group G	Used for	Known uniform samplers	This work
Euclidean	$\Sigma\, x_i\, y_i$	O(n, q) / Ps(n, q)	side channels, SPECK, Hollow LWE	**ITW paper (ℓ = k, ε)**; ABL 2026; q even: none	**all ℓ, all q**
Hermitian	$\Sigma\, x_i\, y_i^q$	U(n, q^2)	stabilizer / EA codes, k − ℓ ebits	none	**all ℓ, all q**
symplectic	a·b′ − b·a′	Sp(2n, q)	stabilizer / EA codes, (k−ℓ)/2 ebits	ℓ = k, q = 2: random Clifford	**all ℓ, all q**

This work: exact, all three forms, every ℓ, every q, expected $O(k^2 N + k \log q)$ per sample.

Why sample uniformly?

quantum codes with a given number of ebits · classification · key generation

	Codes	What ℓ means there	This sampler gives	Before
EA stabilizer codes	symplectic (F_q^{2n}) Hermitian ($F_{q^2}^n$)	$c = (k - \ell)/2$ ebits (Herm.: $k - \ell$) $K = n - (k + \ell)/2$ encoded qudits	uniform EA codes with prescribed (K, c) every c, every q	c = 0 only (random Clifford)
Classification	any form, any ℓ	$P([C]) = n! / (\lvert PAut(C)\rvert \cdot \lvert C_{n,k,\ell}\rvert)$	$T \geq p_{min}^{-1} \ln(M/\delta)$ draws meet all M classes mass formula certifies completeness	enumeration in a fixed order (late classes never reached)
Code-based crypto	Euclidean: $\ell = k$ (SPECK) any ℓ (Hollow LWE)	support splitting costs q^ℓ , so schemes want ℓ large	exact key generation, every characteristic $O(k^2 n + k \log q)$	$\ell = k$: ε / experimental q even: none

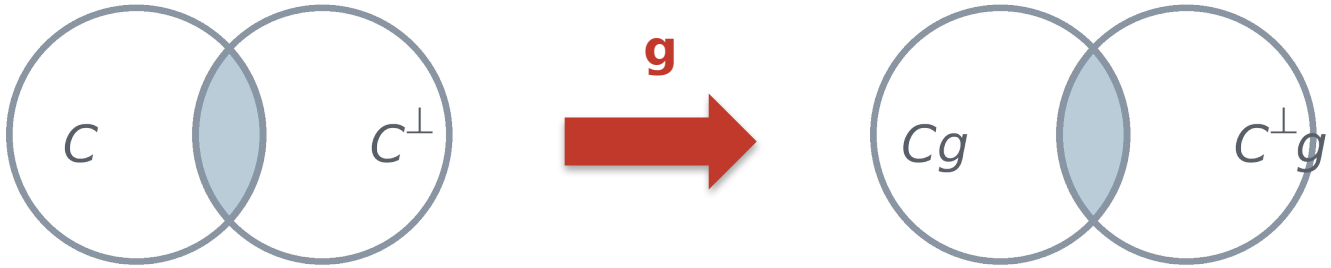
Three communities, one need: a uniform code with prescribed hull dimension.

Isometries: the linear maps that respect $\perp$

From here on: Euclidean inner product, q odd. (The other forms work the same way.)

$G = O(n, q) := \{ g \in GL_n(F_q) : \; g\, g^T = I \}$ (orthogonal group)

Hermitian: unitary group · symplectic: symplectic group · q even: pseudo-symplectic group



g keeps dim C = k and dim Hull(C) = ℓ

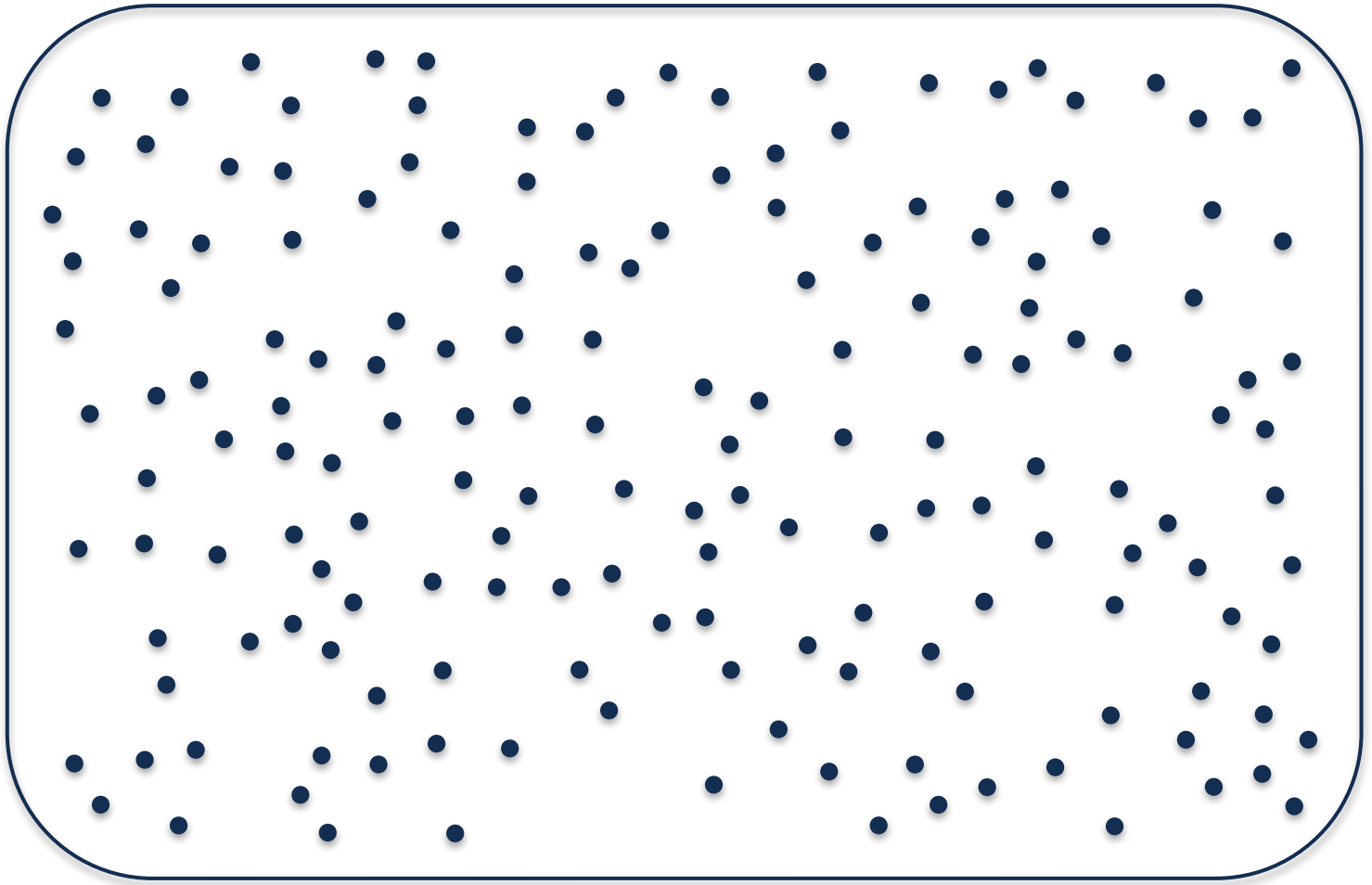
$\Rightarrow$ G acts on $C_{n,k,\ell}$

$(Cg)^\perp = C^\perp g$, Hull(Cg) = Hull(C) g

G permutes the codes in $C_{n,k,\ell}$. Let us draw the picture.

$C_{n,k,\ell}$ is a finite set of codes

Every dot is a code. We want one dot, uniformly at random.



$C_{n,k,\ell}$ = all [n, k] codes with hull dimension ℓ

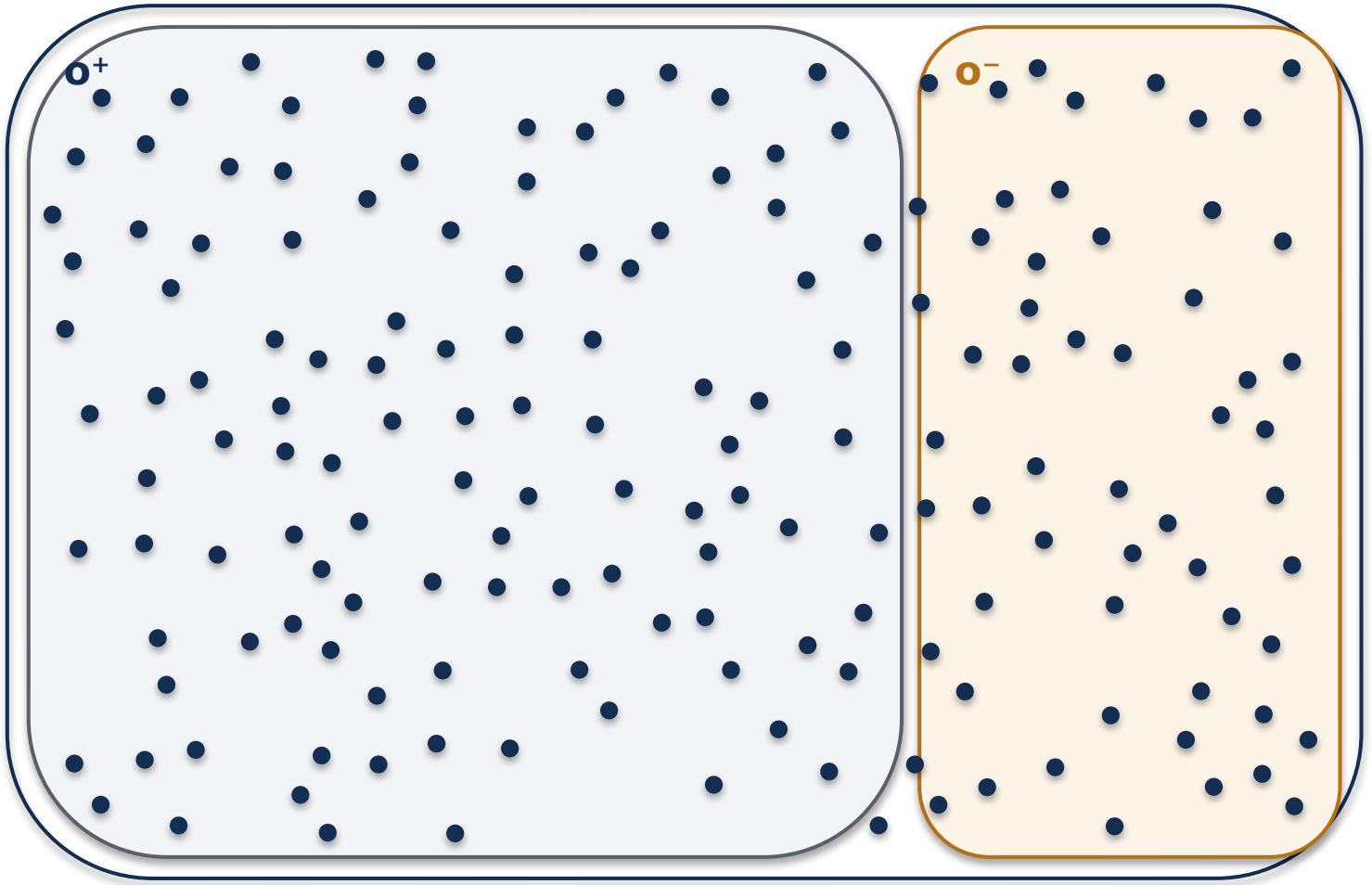
Directly: hard.

- random generator matrix → wrong ℓ, almost surely
- grow the hull one vector at a time → uniformity lost

No structure visible yet.

…but O(n, q) acts on it

One can prove: at most two orbits, of different sizes



$C_{n,k,\ell}$ = all [n, k] codes with hull dimension ℓ

Orbits of O(n, q)

orbit of C ↔ isometry class of C / Hull(C)

= its discriminant, a square or not: $\varepsilon = \pm 1$

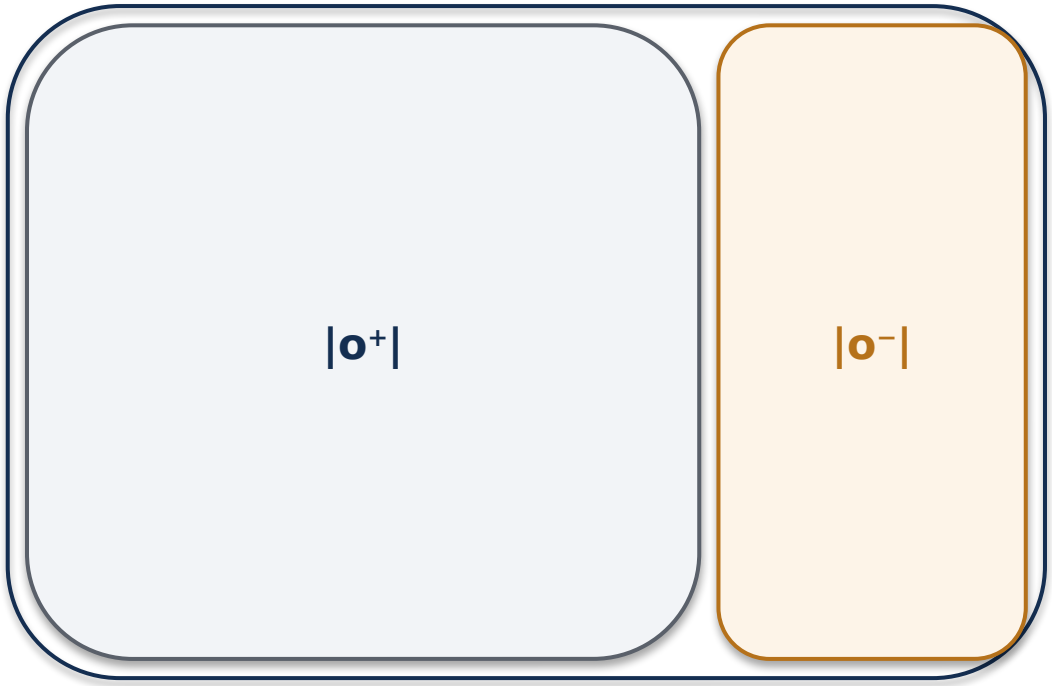
≤ 2 orbits o^+, o^-

other forms: 1 orbit (Hermitian, symplectic), ≤ 4 (q even)

Sample each orbit, weight by size.

Orbit sizes are known ⇒ weighting by size gives uniformity

Anzahl theorems, alias mass formulas, give |o|; then one line of probability



1. pick o with $p^o = |o| / |C_{n,k,\ell}|$
2. draw C uniform in o

If we can draw uniformly inside any orbit:

$$P(C) \;=\; p^o \cdot 1/|o| \;=\; 1 / |C_{n,k,\ell}| \quad \text{for every } C$$

exactly uniform

|o| is known

mass formula of Li–Shi–Ling (2025); in general: Anzahl theorems (Wan). Only the ratio $p^+ : p^-$ is used.

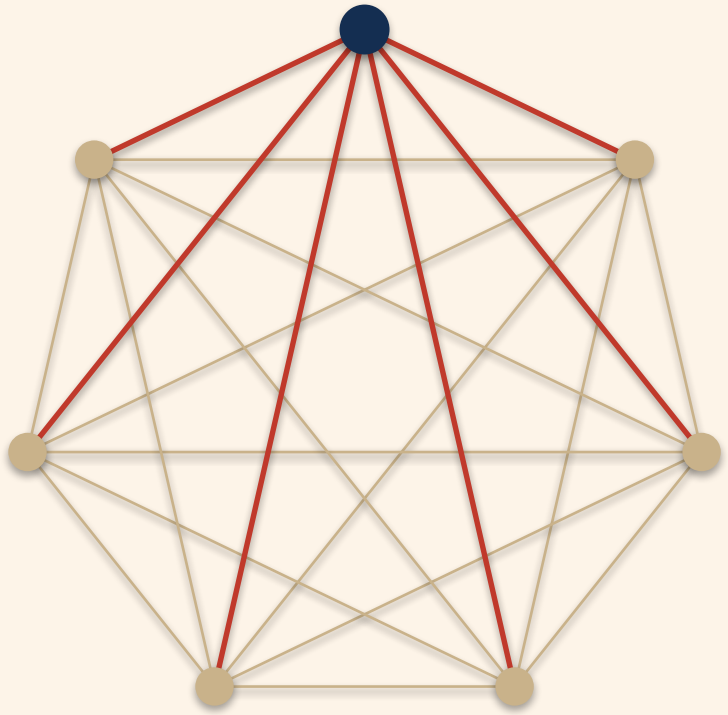
Everything reduces to one orbit. Zoom in.

Inside one orbit: fixed G_0 × uniform g

Transitive ⇒ orbit = { C_0 g : g ∈ G }, and a uniform g gives a uniform C_0 g

orbit o

C_0 = rowsp(G_0)



any two codes are joined by some g; red: C_0 g

fixed
G_0
a simple generator
matrix of C_0

×

uniform
g ∈ G
drawn once
per sample

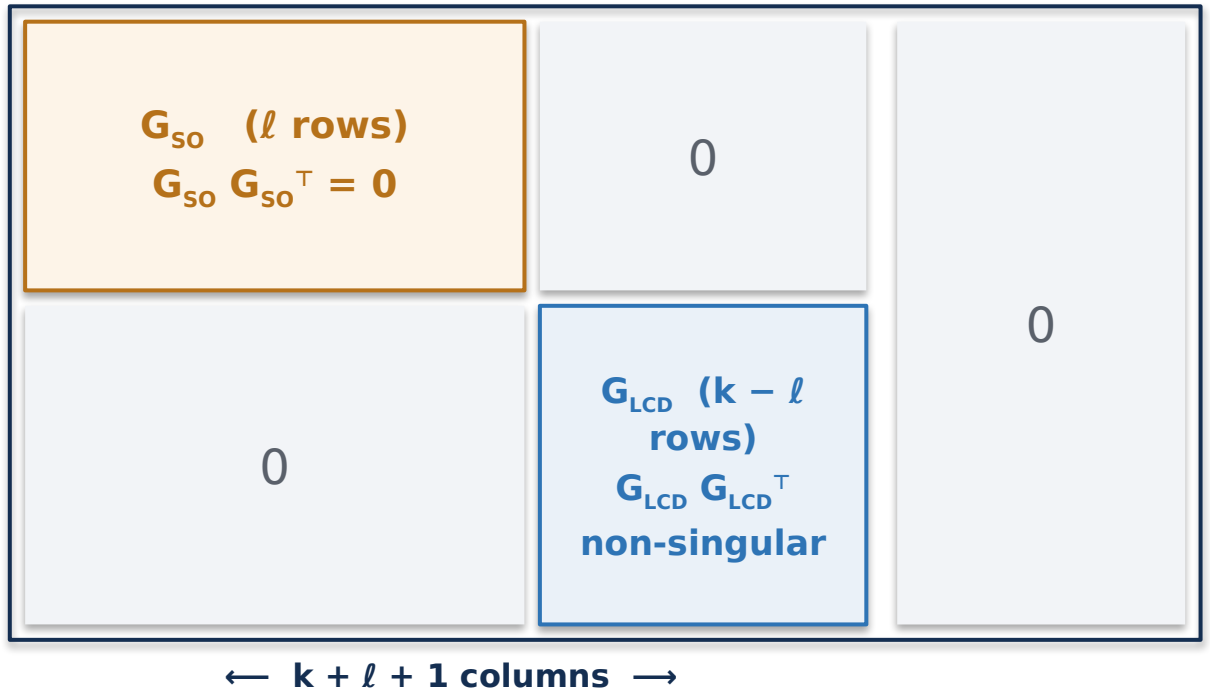
C = rowsp(G_0 g) is uniform on the orbit

Proof (orbit–stabilizer). C_0 g = C_0 h ⇔ g h^{-1} ∈ Stab(C_0),
so every C in the orbit has exactly |Stab(C_0)| = |G| / |orbit| preimages:
P(C) = 1 / |orbit|.

Two problems remain: (1) a simple G_0 for every n, k, ℓ (2) a uniform g ∈ O(n, q)

(1) The representative: self-orthogonal ⊥ LCD

C = Hull(C) ⊥ W with Hull(C) self-orthogonal and W LCD — so $G_0 = \mathrm{diag}(G_{SO}, G_{LCD})$



Gram matrix = $0_\ell \oplus$ (non-singular) ⇒ hull = the first ℓ rows

the self-orthogonal block is the difficult part; the LCD block is just I (or I with one row (a′ b′), a′² + b′² non-square, for the other orbit)

The self-orthogonal block, by q mod 4

q ≡ 1 (mod 4): θ² = −1 has a solution

(1 θ)·(1 θ) = 1 + θ² = 0, so stack rows (1 θ):

$$G_{\mathrm{SO}} = \left(I_\ell \,\middle|\, \theta I_\ell \right)$$

q ≡ 3 (mod 4): no θ, but a² + b² = −1 has a solution

(1 a b)·(1 a b) = 1 + a² + b² = 0, so build rows from (1 a b):

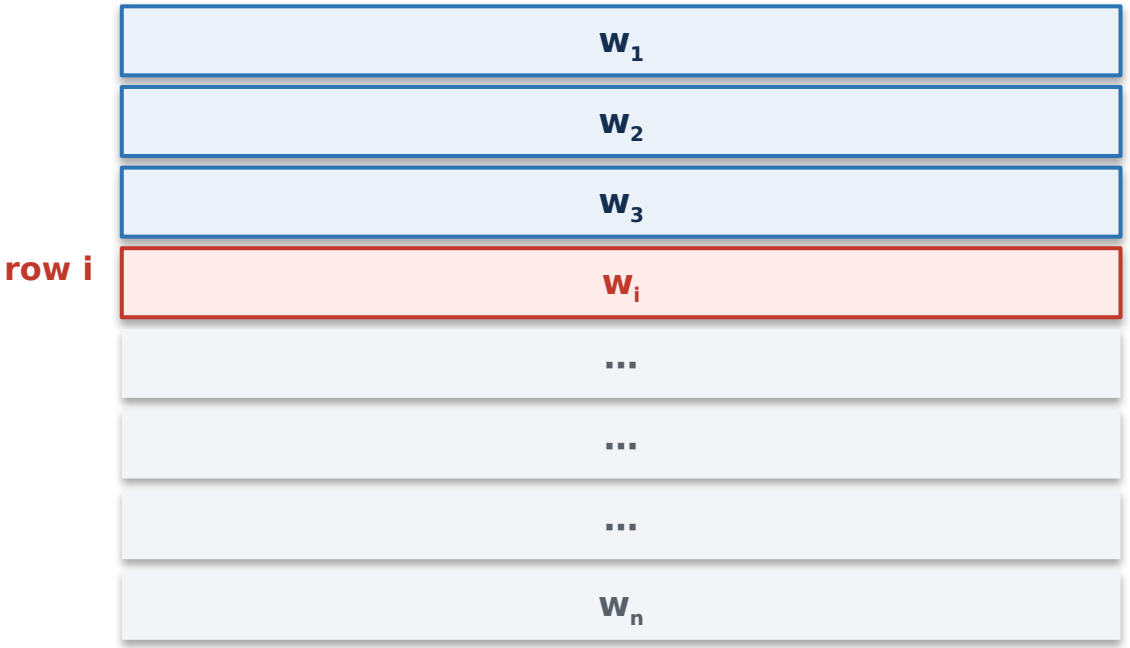
$$G_{\mathrm{SO}} = \mathrm{diag}(M, \ldots, M\,[, r]), \quad M = \begin{pmatrix} 1 & 0 & a & b \\ 0 & 1 & -b & a \end{pmatrix}, \; r = (1\ a\ b)$$

M $M^{\top}$ = 0, r $r^{\top}$ = 0; one r only when ℓ is odd

One recipe for every (n, k, ℓ); it touches only the first k + ℓ + 1 coordinates.

(2) A uniform g ∈ O(n, q): one row at a time

Subgroup algorithm (Diaconis–Shahshahani)



g ∈ O(n, q): rows w_i orthonormal

row i: a uniform unit vector ⊥ w_1, …, w_{i-1}

- draw x uniformly in $\langle w_1, \ldots, w_{i-1}\rangle^{\perp}$
- repeat until x·x is a non-zero square (expected O(1) tries)

One can prove: the number of unit vectors in $\langle w_1, \ldots, w_{i-1}\rangle^{\perp}$ does not depend on the w's

⇒ every g ∈ O(n, q) equally likely.

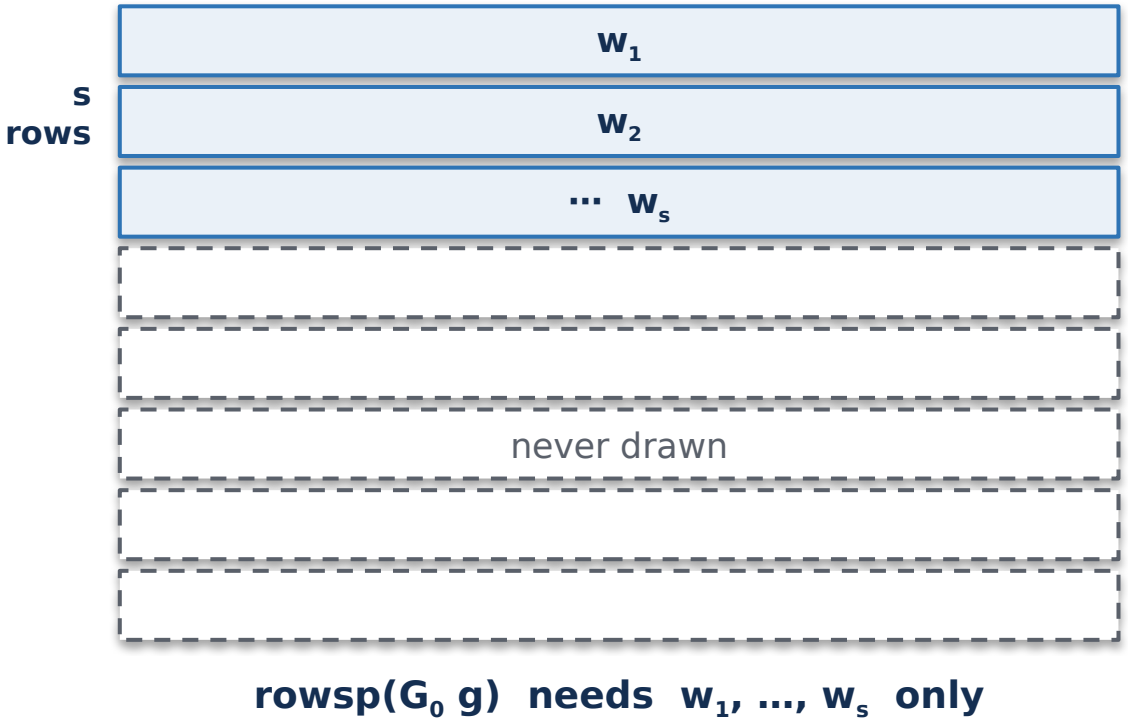
row i: linear system in echelon form, O(in)

all n rows: **$O(n^3)$**

Exactly uniform g in $O(n^3)$ ⇒ exactly uniform C in $O(n^3)$.

…and stop after the first s = k + ℓ + 1 rows

G_0 uses the first s coordinates only, so G_0 g uses only w_1, …, w_s



rowsp(G_0 g) needs w_1, …, w_s only

allowed: the first s rows have the same law whether or not we continue

pays: $\Sigma_{i \le s}$ O(in) = O(s^2 n) = **O(k^2 n)**, + O(k log q) for the square roots

Main theorem. Exactly uniform codes with prescribed hull dimension,
all three forms, every q, expected **O(k^2 N + k log q)** per sample
(N = dim V: n, or 2n for the symplectic form).

ITW paper (ℓ = k): ε-uniform, O(n^5 log q). Other forms, q even, proofs: the paper.

Exact uniformity for the price of a k × N matrix, not an N × N one.

Summary

Uniform sampling of codes with prescribed hull dimension

1 $C_{n,k,\ell}$ = at most four orbits of the isometry group; sizes = mass formulas

2 one fixed representative + one uniform g ⇒ exactly uniform code

3 g row by row (exact, $O(N^3)$), stop after $k + \ell + O(1)$ rows ⇒ $O(k^2 N + k \log q)$

4 EA codes with given ebits · classification coverage · SPECK / Hollow LWE

ITW paper: Euclidean, q odd, $\ell = k$, ε-uniform. Journal version: arXiv:2611.03578.

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Thank you — questions?